

Draft results section for collaborator review

Historical NGC 5548 campaigns 7 October 2026

This preview contains the completed descriptive analysis. The bounded categorical calibration and ordering audit are complete. The analysis is frozen for review; observation matched calibration and expanded scientific runs remain unperformed.

1 Campaign resolved SURD components and sampling sensitivity

1.1 Observed series and analysis scope

We analyze historical observations of NGC 5548 in five observing campaigns from 1989 through 1993. The adopted line sample contains 242 spectra on 224 distinct dates. Corrected published total, blue, core, and red $H\beta$ measurements are used directly from the table accompanying Wanders and Peterson (1996), together with 557 original optical continuum measurements at 5100 Å. Duplicate spectral dates are represented by equal means, with mean uncertainties obtained from the quadrature sum of quoted errors divided by the number of measurements. Shared calibration covariance is unavailable. The original continuum retains host starlight. All four line variables share spectra, and the total nearly equals the sum of the three components. We retain the published rounding and do not treat these measurements as independent observations of four physical regions.

Campaign membership follows the intervals in Table 2 of Peterson et al. (2004): JD less 2400000 from 47509 through 47809, 47861 through 48179, 48225 through 48534, 48623 through 48898, and 48954 through 49255. The blue, core, and red definitions follow the observed wavelength intervals in source Section 3.2, respectively 4870 through 4920, 4920 through 4960, and 4960 through 5010 Å. The source figure caption gives a different blue lower boundary; using the published measurements avoids claiming an exact reconstruction of the integrations from the archived profile pixels.

For each campaign, we analyze two variable sets: total, blue, core, and red $H\beta$; and continuum, blue, core, and red $H\beta$. Every variable in a set becomes the future target in turn, while all four past variables, including the target’s own history, remain predictors. We use the unchanged SURD implementation of Martínez Sánchez et al. (2024), with the averaging and normalization described by Martínez Sánchez (2025). For four predictors, the listed combinations give four unique, eleven redundant, and eleven synergistic components, hence 26 in total.

For future target $Y(t)$ and predictor vector $\mathbf{X}_\tau(t)$, we write each normalized component as

$$f_a(\tau) = \frac{I_a(Y; \mathbf{X}_\tau)}{I(Y; \mathbf{X}_\tau)}, \quad \sum_{a=1}^{26} f_a(\tau) = 1, \quad (1)$$

when joint mutual information is nonzero. Information leakage uses a separate denominator,

$$\ell(\tau) = \frac{H(Y | \mathbf{X}_\tau)}{H(Y)}. \quad (2)$$

The component fractions therefore describe the allocation of available predictive information, while leakage describes the fraction of target entropy left unexplained. A large normalized component can coexist with substantial leakage.

The saved exploratory scans cover base lags from zero through 30 observed days. Predictors are interpolated at native target dates within the same campaign, with no extrapolation and a maximum interpolation gap of ten days. We retain a minimum of 32 finite tuples as a numerical guard. Probability histograms use either two or three marginal quantile bins, whose boundaries are determined from each full native campaign series. The two additional scenarios place the blue or red predictor five days earlier than the common base lag. These stipulated offsets are sensitivity choices, not measured delays. Their construction gives one lag to each predictor; it does not execute the expanded predictor vector of thesis equation 3.18.

1.2 Campaign support and sensitivity to bin choice

Each bin setting contains 3720 configurations, with 1402 passing the numerical guard. At positive lags there are 1326 supported configurations. The native inputs, settings apart from bin count, tuple counts, and support status agree between the two scans. This permits a direct comparison of bin settings at each lag. It does not hold target dates fixed across lags.

We quantify the change in the normalized component distribution using

$$D_{\text{TV}}(\tau) = \frac{1}{2} \sum_{a=1}^{26} \left| f_a^{(2)}(\tau) - f_a^{(3)}(\tau) \right|. \quad (3)$$

The superscripts denote bin count. Leakage changes are reported as $|\ell^{(2)} - \ell^{(3)}|$. Both quantities are descriptive diagnostics; neither is a significance test or an uncertainty estimate.

Table 1: Native date counts and positive lag support by campaign. Each campaign has 720 positive lag configurations per bin setting. N_{line} is the number of native dates for each line series and N_F is the continuum count. N_{usable} applies to the earlier support that changes with lag. The last column gives the range of retained date counts after intersection across all positive lags in each curve.

Campaign	N_{line}	N_F	N_{usable}	Median D_{TV}	Fixed dates
1989	48	124	111	0.466	1 to 7
1990	38	94	68	0.574	1 to 7
1991	32	65	1	0.474	0 to 0
1992	46	83	426	0.403	22 to 36
1993	60	191	720	0.515	25 to 90

Table 1 shows that numerical support differs strongly between campaigns. Only one positive lag configuration is usable in 1991. All 720 configurations in 1993 pass the guard, but the median D_{TV} is 0.515 and every supported configuration in that campaign has $D_{\text{TV}} > 0.2$. Across all campaigns, median D_{TV} is 0.484 and median absolute leakage change is 0.182. Thus even the campaign with complete numerical coverage is sensitive to the probability discretization. Complete coverage under the tuple guard does not establish reliable estimation of the component fractions.

1.3 Holding target dates fixed across all positive lags

To remove changes in native target support along a lag curve, we retain only target dates at which every required predictor is finite at every base lag from one through 30 days. This intersection is computed separately for each campaign, variable set, target, and scenario. The resulting dates are used at every lag and for both bin settings. The lag range, interpolation gap limit, numerical guard, and full campaign quantile boundaries are unchanged. Zero lag is excluded from the intersection and comparison.

Only four of 120 curves retain at least 32 target dates: the common continuum target in 1992, and the common, blue earlier, and red earlier continuum targets in 1993. Every future line target fails the guard under this stricter construction. The 116 excluded curves have no decomposition in this run. The four retained curves produce 240 decompositions across 30 lags and two bin settings, with 6240 component rows. Normalization is defined in all executed cases, with maximum absolute component sum error 1.22×10^{-15} ; this verifies arithmetic only.

Table 2: Sensitivity between bin settings with identical target dates across all positive lags. Every row has future continuum as the target. Each median uses 30 base lags. The last column counts components whose maximizing lag sets do not overlap between bin settings. All components, including weak ones, are counted; ties are retained within absolute tolerance 10^{-12} .

Campaign	Scenario	Dates	Median D_{TV}	Median $ \Delta\ell $	Disjoint peaks
1992	Common	36	0.531	0.125	22/26
1993	Blue earlier	78	0.610	0.144	25/26
1993	Common	90	0.594	0.122	21/26
1993	Red earlier	78	0.523	0.170	21/26

The sensitivity persists after fixing the target dates (Table 2). Median D_{TV} ranges from 0.523 to 0.610. In addition, changing the retained observations affects the estimates at a fixed bin count. For the 1993 common curve, median D_{TV} between the earlier varying support estimates and the new fixed support estimates is 0.475 for two bins and 0.391 for three bins. Across the four retained curves and both bin settings, these medians range from 0.272 to 0.475. This measures sensitivity to sample selection and can include temporal variation; it does not isolate estimator bias.

The retained sample has 36 dates for the 1992 curve and 78 through 90 dates for the 1993 curves. In the 1993 common case, median occupied histogram cell counts are 14.5 for two bins and 25 for three bins, with 90 tuples. The corresponding medians of singleton cell counts are four and ten. Repeated or interpolated observations are not independent samples, and structural dependence can also leave cells empty. These occupancy counts are diagnostics rather than estimator validation.

1.4 Individual components and descriptive extrema

Figures 1 through 4 show all individual components and leakage for the four retained curves. The notation uses F for continuum, B for blue, C for core, and R for red; the outer operator identifies a unique, redundant, or synergistic component. Dotted curves use two bins and solid curves use three bins. All component panels share the joint information denominator in equation 1, while the leakage panel uses equation 2.

The 1993 common case is selected as the example with the largest retained date count, rather than by the amplitude or location of a peak. Table 3 reports the largest component value within each information type over the declared positive lag range. The largest unique contribution comes from continuum history in both bin settings, but its maximizing lag changes from six days to one day. The largest redundancy changes from $R(B, C)$ at 25 days to $R(B, C, R)$ at eleven days. The largest synergy changes from $S(F, R)$ at the 30 day scan boundary to $S(F, C)$ at four days. The boundary maximum cannot identify an interior characteristic delay. Component identities, amplitudes, and maximizing lags therefore change with discretization despite identical native target support.

Table 3: Descriptive maxima in the 1993 common scenario with 90 fixed continuum target dates. Each row maximizes over all components of the specified type and all base lags from one through 30 days. These maxima have no null calibration or delay uncertainty and are not physical lag measurements.

Bins	Type	Component	Base lag (days)	Maximum fraction
2	U	U(F)	6	0.636
2	R	R(B,C)	25	0.381
2	S	S(F,R)	30	0.244
3	U	U(F)	1	0.547
3	R	R(B,C,R)	11	0.244
3	S	S(F,C)	4	0.181

1.5 Ideal categorical estimator calibration

We check the same four predictor estimator against eight exact categorical laws: independent variables, a target equal to one predictor, a target duplicated by two predictors, and a target given by the modular sum of two independent predictors, each with two or three states. The dependent laws have known unique, redundant, or synergistic information, respectively. Their exact joint information is the base two logarithm of state count and their leakage is zero. The independent law has zero joint information and unit leakage, so its exact normalized component fractions are undefined. The official routine reproduces these exact reference values in the declared predictor order.

We draw 200 independent samples per law and sample size with fixed seed 20261007, giving 6400 decompositions. Sizes of 36, 78, and 90 match the retained curve counts; 512 is an artificial reference. Table 4 shows substantial apparent information under independence, particularly with three states. This benchmark demonstrates finite sample bias and does not provide an AGN significance threshold. These are separate categorical laws, not rebinning of one continuous dataset. They omit temporal dependence, interpolation, measurement errors, and fitted quantile boundaries. The deterministic positive controls also do not represent weak physical signals.

Table 4: Median joint information I in bits and normalized leakage ℓ under independent categorical variables, from 200 draws per setting. Exact truth is $I = 0$ and $\ell = 1$. The 512 tuple setting is an artificial reference. None of these values is a significance threshold for the AGN.

Tuples	I , two states	I , three states	ℓ , two states	ℓ , three states
36	0.353	1.304	0.641	0.157
78	0.158	1.076	0.839	0.313
90	0.136	1.015	0.862	0.353
512	0.019	0.268	0.981	0.830

Some binary positive controls also show unexpected component allocations. In four preserved examples, reordering predictor histogram axes together with their physical labels changes the normalized allocation substantially while leaving total information unchanged. An execution trace of the unchanged routine records resets to zero of positive specific information after strict comparison differences around machine precision. We assess the relevance of this numerical concern to the retained observations next.

1.6 Predictor ordering audit of the retained observations

For every retained curve, bin setting, and positive lag, we reconstruct the original histogram and verify reproduction of the saved components to absolute tolerance 10^{-10} . Target dates and tuple counts also agree. We then evaluate all 24 orders of the four predictor axes, moving their names with their axes and keeping the target axis fixed. Components are matched by type and the set of named predictors. There are 5760 order evaluations across 240 lag and bin configurations. The histogram counts, target dates, bin boundaries, lag range, and algorithm are unchanged.

An allocation is flagged when its total variation from the original order exceeds 10^{-8} , a declared numerical comparison tolerance rather than a significance threshold. Eleven configurations exceed this tolerance (Table 5). The largest total variation is 0.188 in the 1993 red earlier scenario with three bins. Maximum absolute changes in joint information and leakage across the audit are 6.66×10^{-16} bits and 1.83×10^{-15} , respectively. Thus stable total information and component sum normalization do not ensure stable allocation to named components.

Table 5: Predictor ordering sensitivity of the retained astronomical histograms. Each row contains 30 positive lag configurations, each checked in all 24 predictor orders. Affected lags have component total variation above 10^{-8} in at least one order. The final column is the maximum over all lags and orders in that row.

Campaign	Scenario	Bins	Affected lags	Maximum D_{TV}
1992	Common	2	5/30	0.065
1992	Common	3	3/30	0.040
1993	Blue earlier	2	2/30	0.107
1993	Blue earlier	3	0/30	$< 10^{-8}$
1993	Common	2	0/30	$< 10^{-8}$
1993	Common	3	0/30	$< 10^{-8}$
1993	Red earlier	2	0/30	$< 10^{-8}$
1993	Red earlier	3	1/30	0.188

Both 1993 common curves are stable to predictor ordering within this tolerance. Their strong bin sensitivity therefore cannot be attributed solely to this ordering issue. The figures and extrema retain the original saved ordering as computational diagnostics. We do not select a preferred permutation or alter the authors’ routine to obtain a physical result. The ordering concern is present in a subset of the observed configurations and limits physical interpretation of individual allocations.

1.7 Astrophysical scope and remaining analysis

The executed curves describe future continuum prediction from its own history and past line components. They do not provide a supported fixed date curve of future line response to past continuum in the declared lag range. A predictive association also requires consideration of a shared ionizing driver, measurement noise, temporal dependence, and the imperfect relation between observed optical continuum and illuminating radiation before it can receive a causal interpretation. Total and component line measurements retain their shared spectral provenance.

Lu et al. (2022) report emission line reverberation relative to the optical continuum in five modern seasons and changes in velocity resolved lag profiles. Xi et al. (2025) extend the program to 2023 and report temporal evolution of the inferred gas kinematics. Those studies motivate analysis of separate campaigns. Their line response delays and kinematic interpretations are different quantities from the normalized information component extrema reported here. Our historical epochs, wavelength partitions, observed time frame, and retained target direction also differ. The present diagnostics therefore do not establish agreement or disagreement with their physical delay measurements or with an inflow or outflow model.

The completed calculations establish substantial sensitivity to bin choice and retained dates. Neither bin setting is selected as reliable, and no stable physical delay or causal detection is claimed. No significance test or flux uncertainty propagation has been performed for these fixed support curves. The initial categorical calibration and the complete ordering audit of the retained curves are finished. Calibration matching the temporal and observational properties remains unperformed. The expanded predictor construction of thesis equation 3.18 is implemented and tested, but only its support audit has been executed; an expanded scientific decomposition is not reported. These limitations delimit the present results section and the remaining collaborator analysis. The exploratory analysis is frozen at this stage for collaborator review. Further scientific runs, any change to the authors’ routine, and any revised probability estimation or lag settings require a new agreed analysis scope.

1992: common, continuum target, 36 fixed dates

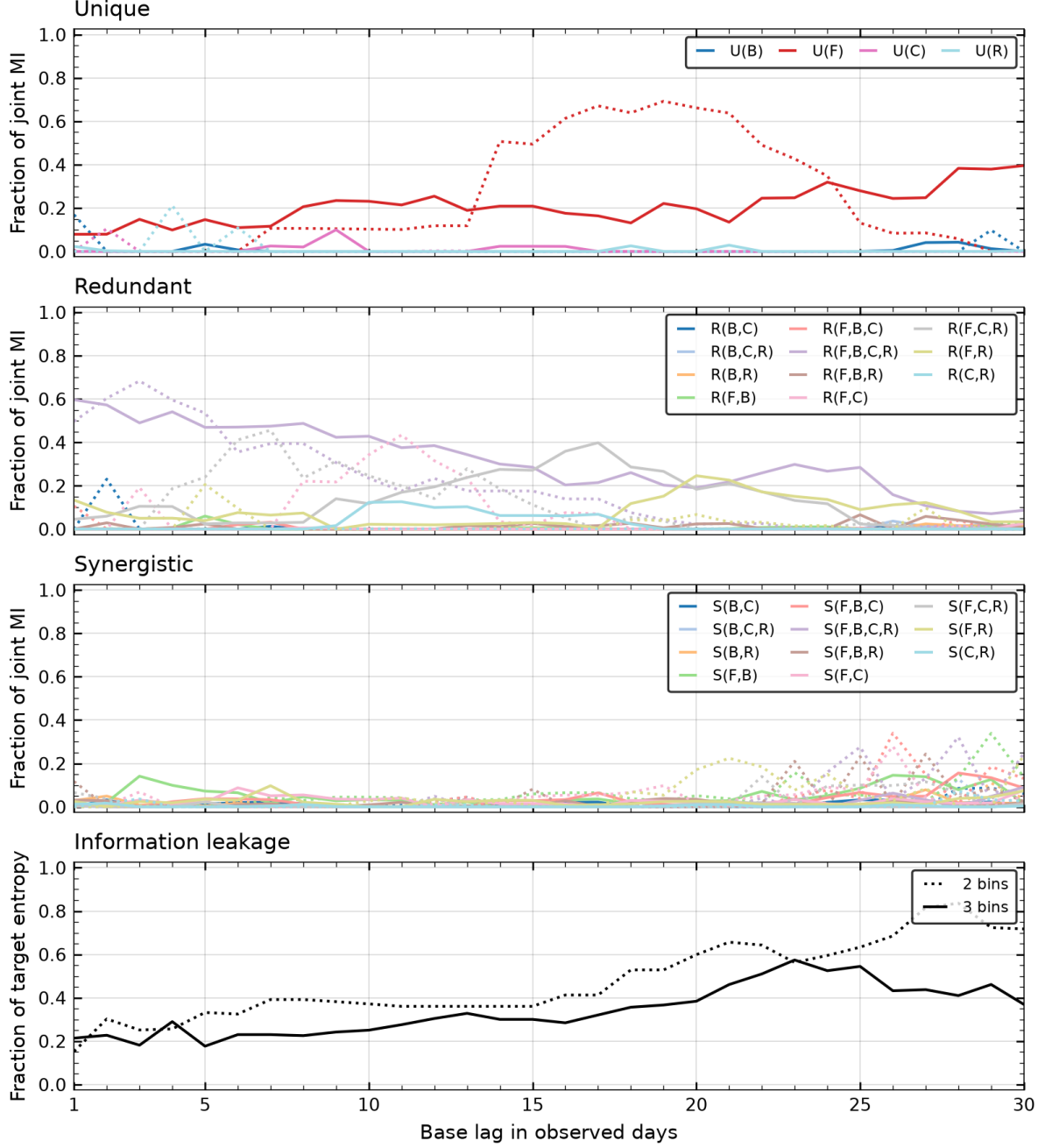


Figure 1: Individual normalized SURD components and information leakage for the 1992 common scenario. The future target is continuum and predictors are past continuum, blue, core, and red $H\beta$. Every base lag uses the same 36 target dates. F , B , C , and R denote those four variables. The outer U , R , and S identify the information type. Dotted curves use two bins and solid curves use three bins. Leakage has its own target entropy denominator. The curves are descriptive estimates without significance or physical delay claims.

1993: common, continuum target, 90 fixed dates

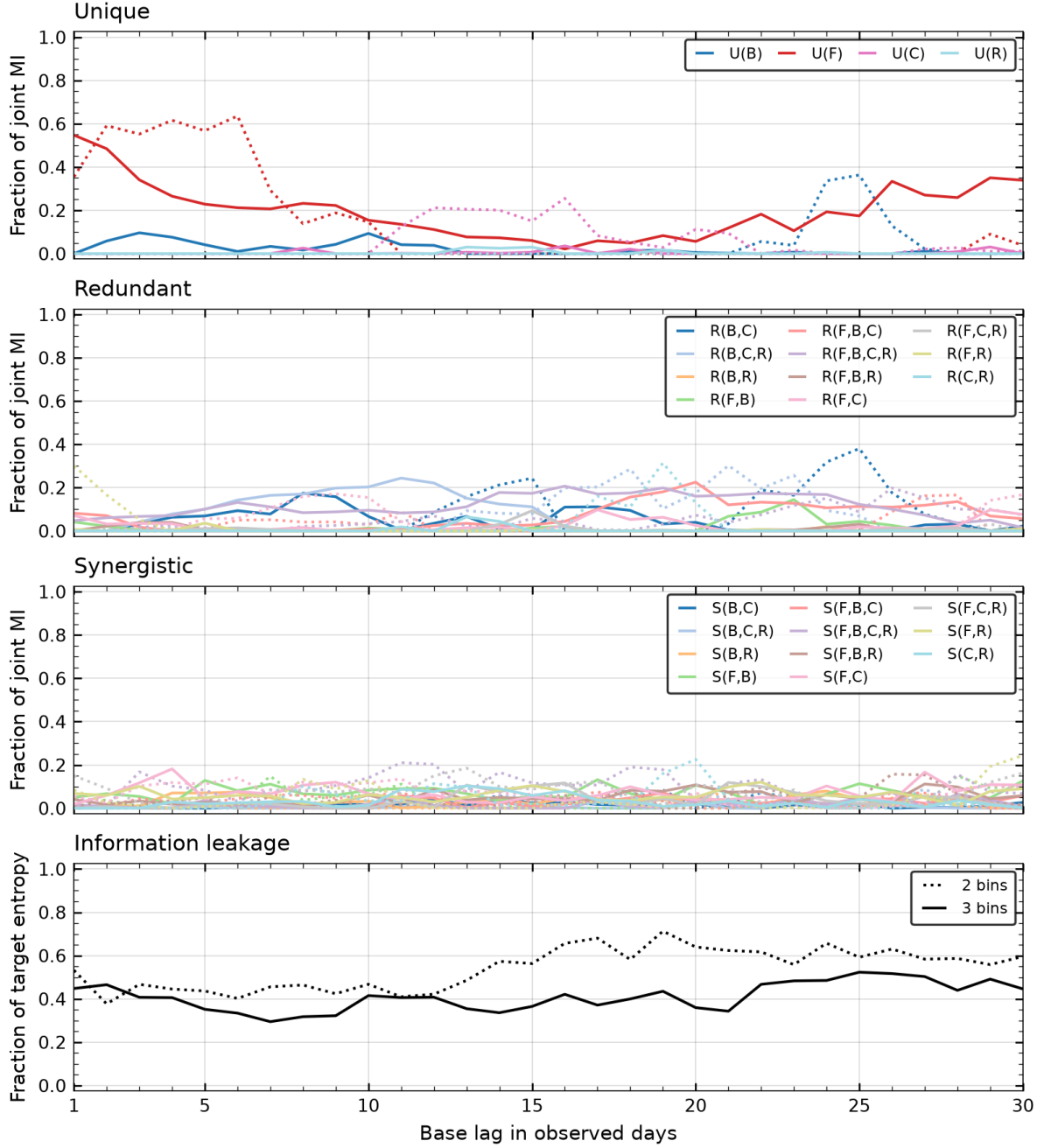


Figure 2: As in Figure 1, for the 1993 common scenario with 90 fixed continuum target dates. This is the retained curve with the largest native target support. The changes in individual maxima between bin settings are summarized in Table 3.

1993: blue earlier, continuum target, 78 fixed dates

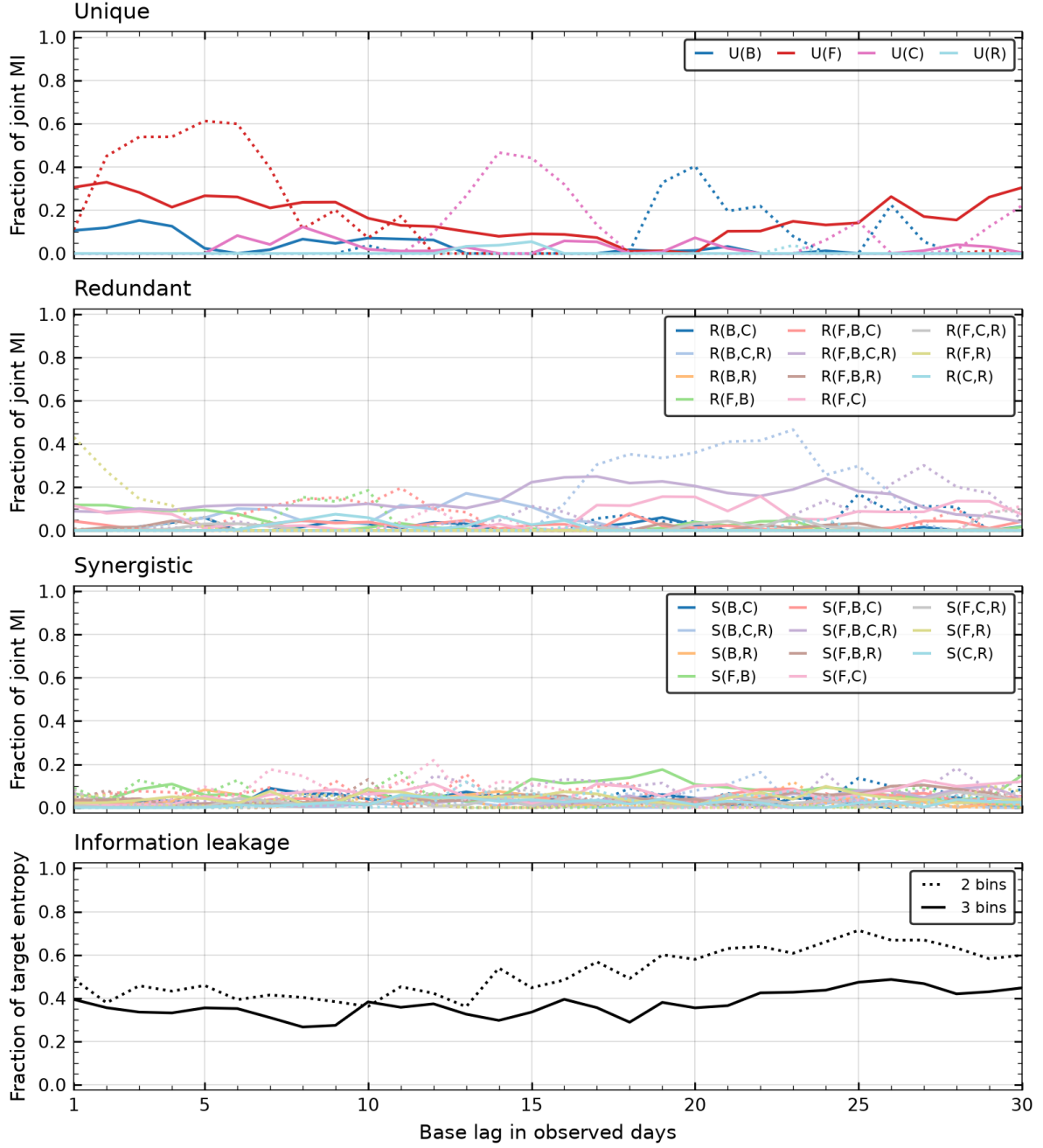


Figure 3: As in Figure 1, for the 1993 blue earlier scenario with 78 fixed continuum target dates. The blue predictor is evaluated five days earlier than the other predictors. This stipulated offset is a sensitivity choice, not a measured delay. Its retained dates differ from those in the common scenario, so scenario differences cannot be attributed solely to the predictor offset.

1993: red earlier, continuum target, 78 fixed dates

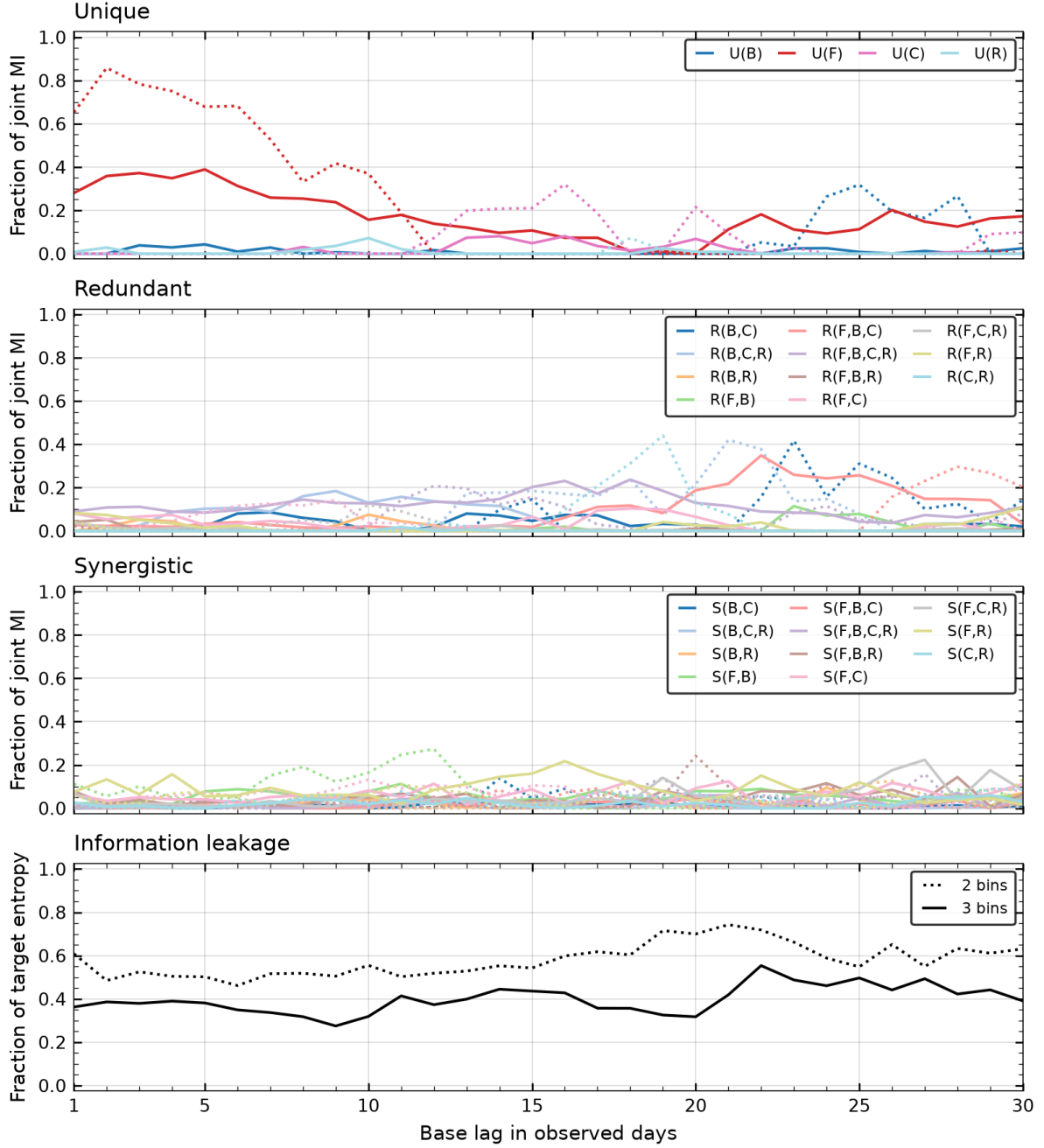


Figure 4: As in Figure 1, for the 1993 red earlier scenario with 78 fixed continuum target dates. The red predictor is evaluated five days earlier than the other predictors. This is a stipulated sensitivity choice. Dates are held fixed within this curve, but are not forced to match the other scenarios.

References

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